Mutual Revision Between Partial Learners: A Pre-Registered Toy Study of Mutual Distillation on Unlabeled Inputs

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Abstract

Can two models that each learned from a different slice of the data end up knowing more by revising their predictions toward each other than either (or both, without interacting) knows alone? We test this in a deliberately small setting: modular addition modulo 53, two small networks that each receive labels for a different part of the data, and a shared pool of unlabeled inputs on which each network learns from the other's predictions (mutual distillation). The hypotheses and decision rules were pre-registered before any run. In the pre-registered regime (18% of all pairs labeled per model), nothing happens: every model, alone or interacting, scores 0% on held-out pairs. In a near-threshold regime added after seeing that null (30% labeled per model), mutual distillation reaches 99.7% test accuracy against 32.7% for a model alone and 43.1% for a non-interacting ensemble, essentially matching a model trained on both label sets pooled (100%). The effect disappears when both models see the same labeled data, is not explained by longer training, and occupies a narrow band of labeled-data fractions (roughly 24–28% per model) that lies below where a single model begins to generalize. Pooled labels are never beaten. The mechanism resembles known co-training and mutual-learning methods; the contribution here is a pre-registered, controlled characterization in a toy setting, with the post-hoc nature of the positive result stated plainly.

1 Introduction

A common intuition is that capability comes from scaling a single system. A different possibility is that some capability comes from *interaction* between systems that learned from different histories: neither system alone has enough to solve a problem, but each holds something the other lacks, and exchanging that, even through a narrow channel, lets both cross a threshold. This note tests a minimal version of that idea. It is a toy test of one mechanism, not a test of any broader claim about intelligence or emergence.

The setting is chosen so that the idea can be tested cleanly. Modular addition has a known data-amount threshold for generalization, the phenomenon known as grokking [Power et al., 2022, Nanda et al., 2023], so a regime where one model has too little data and two together have enough can be constructed and verified. Two small networks each receive labels for a different slice of the input pairs. Both can also see a shared pool of inputs *without labels*, and learn from each other's predictions on it. We compare against controls designed to separate interaction from other explanations: each model alone, an ensemble that never interacts, a model distilling from a lagged copy of itself, pooled labels, a peer with the same labeled data, and a solo model trained four times longer.

The study was pre-registered: hypotheses, conditions and the decision rule were written before any experiment was run [Nosek et al., 2018]. The pre-registered regime produced a null. A second,

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near-threshold regime was then added; its positive result is therefore exploratory, and we report it as such.

Related work. The mechanism studied here is not new. Deep mutual learning trains peer networks that each distill from the others' predictions [Zhang et al., 2018]; codistillation does the same at scale [Anil et al., 2018]; both build on knowledge distillation [Hinton et al., 2015]. Co-training exploits unlabeled data by letting two learners label examples for each other [Blum and Mitchell, 1998]. Our setup is closest to mutual learning applied to unlabeled shared inputs, with the labeled data partitioned between the peers. We do not claim novelty of the method. The contribution is a pre-registered controlled characterization of when it helps in a setting where the answer can be checked, including a clear null.

2 Experimental Setup

Task and data. The task is $y = (a + b) \bmod 53$ over all $53 \times 53 = 2809$ input pairs, each encoded as the concatenation of two one-hot vectors. For every seed the pairs are divided into five disjoint pools: two labeled pools S_A and S_B (one per model), an unlabeled pool U (inputs only, shared by both models), a validation set (hyperparameter selection only), and a test set that is never used for training or tuning. Two split modes are used. In the *random* mode, S_A and S_B are random disjoint subsets. In the *region* mode, S_A contains only pairs with $a < 27$ and S_B only $a \geq 27$, so each model has seen labels from one part of the input space only, and accuracy can be reported per region. Table 1 gives the pool sizes. A script checks that all pools are disjoint and that the region mode respects its regions for every seed used.

Table 1: Regimes and pool sizes (number of pairs out of 2809). Regime P is pre-registered; regime T was added after P produced a null.

Regime	Labeled / model	S_A	S_B	U	Validation	Test
P (pre-registered)	18%	505	505	674	280	845
T (post hoc)	30%	842	842	561	224	340

Models. Each model is a two-layer perceptron, one-hot$(106) \to$ Linear$(H) \to x \mapsto x^2 \to$ Linear(53), written in NumPy. The "small" model has $H = 128$; the "big" model used for one pooled control has $H = 512$.

Training. All models are trained full-batch for 6000 steps with AdamW (learning rate 3×10^{-3}, decoupled weight decay 1.0, $\beta = (0.9, 0.98)$). The big pooled model uses weight decay 0.5. Training hyperparameters were tuned on validation data using only the solo and pooled conditions, on one seed, and frozen before any interaction condition was run. Metrics are test accuracy at the final training step; there is no early stopping on test.

Mutual distillation. Model i minimizes the cross-entropy on its own labeled pool plus a distillation term on the unlabeled inputs,

$$L_i = \mathrm{CE}(f_i(x), y)_{(x,y)\in S_i} \; + \; \lambda(t)\, \frac{1}{|U|} \sum_{x \in U} w(x)\, \mathrm{KL}(p_j(x) \,\|\, p_i(x))\,, \tag{1}$$

where p_i, p_j are the softmax outputs of model i and its peer j, the peer's output is treated as a fixed target, and $\lambda(t)$ is zero during a warm-up and then ramps linearly to its final value over 500 steps. No labels from U, validation or test are used. The weight $w(x)$ defines three variants. *Uniform*: $w = 1$. *Challenge-weighted*: $w \propto \mathrm{JS}\big(p_i(x), p_j(x)\big)$, emphasizing inputs where the models disagree. *Agreement-weighted*: $w \propto 1 - \mathrm{JS}\big(p_i(x), p_j(x)\big) / \ln 2$, emphasizing inputs where they agree. Weights are normalized to mean 1 over U and clipped at 10. The two mutual models start from the same initial weights as the corresponding solo models, so comparisons are paired within a seed.

Conditions. Table 2 lists every condition and what it controls for.

Interaction hyperparameters. The final distillation weight λ and warm-up length were chosen per condition on validation data, on a separate tuning seed, from a small grid, and then frozen. In regime T the chosen settings were λ=3.0, warm-up 300 for uniform mutual; λ=1.0, warm-up 300

Table 2: Conditions and what each controls for.

Condition	Description / purpose
solo	Each model trained on its own labels only (baseline).
ensemble	Average of solo A and solo B predictions, no interaction.
pooled small / big	One model (H=128 / H=512) trained on $S_A \cup S_B$ labels: the strongest label-only reference.
self-teacher	Each model also distills, on U, from a lagged copy of itself (refreshed every 100 steps). Controls for gains from merely seeing the inputs in U.
mutual (uniform / agreement / challenge)	The interaction conditions above.
same-data peer	Both models get the *same* labeled set S_A (only the initialization differs) and run the uniform mutual procedure. Tests whether a different data history is required.
solo, 4× longer	Solo training for 24000 steps. Tests whether the effect is only extra compute.

for agreement- and challenge-weighted; and λ=1.0, warm-up 1500 for the self-teacher. In regime P every setting scored 0% on validation, so λ=1.0, warm-up 300 was used throughout. Because settings were tuned separately per condition, small differences between mutual variants can reflect tuning rather than the weighting itself.

Statistics and decision rule. Results are means over seeds (4 for regime P, 6 for regime T, 3 for the sweep), with paired comparisons within seed and 95% percentile bootstrap confidence intervals over seeds [Efron, 1979]. A difference counts as *real* only if its interval excludes zero *and* the mean gap is at least 3 accuracy points. With so few seeds the intervals are crude.

3 Pre-Registration and Validation

The pre-registration fixed five hypotheses: **H1**, mutual and challenge-weighted learning beat solo, ensemble and self-teacher; **H2**, challenge-weighting beats uniform and agreement-weighting; **H3**, the best interacting condition matches or beats the big pooled model (prior probability stated as about 10%); **H4**, interaction reduces diversity between the two models (CKA up, prediction disagreement down); **H5**, interaction does not reduce each model's accuracy on its own region by more than 3 points.

Three checks support the correctness of the pipeline. (i) The split script above confirms there is no overlap between pools. (ii) To confirm that the null in regime P is not a bug in the interaction code, an oracle diagnostic gave each model a peer with perfect knowledge of the labels on U; the models then reached 86–98% validation accuracy in that regime, so the interaction machinery does learn when the teacher is informative. (iii) A PyTorch port reproduces the NumPy training loop: from identical weights and data, after 150 full-batch steps the final logits agree to about 5×10^{-5} and predictions match on 100% of test inputs for all six training conditions. One full seed of regime T rerun end to end in PyTorch gave solo 29.4% (NumPy 29.0%), uniform mutual 99.7% (99.0%) and challenge-weighted 100% (100%). All reported results come from the NumPy code.

Deviations from the pre-registration. (1) Regime T was added after regime P gave a null; P remains the pre-registered result. (2) The same-data control, the 4×-longer solo control, the oracle diagnostic and the labeled-fraction sweep were added after seeing results, to rule out explanations suggested by them. (3) Interaction hyperparameters in regime T were tuned after the null in regime P. (4) "Matches or beats" in H3 was read as "within 3 points counts as a tie." (5) Seeds are few, and the sweep uses new seeds (10–12) with settings tuned only at 30% labeled.

4 Results

4.1 Pre-Registered Regime: a Null

At 18% labeled per model (regime P, 4 seeds), no model generalizes. Every solo model, the ensemble, the self-teacher and all three mutual variants score 0.0% on test, with identical per-seed ranges. Pooling the labels does generalize (74.7% for the small model, range 67.1–85.7%, and 93.2% for the big model), so the task is learnable at this data amount; the individual models simply have too little to start from, and two models that know nothing cannot teach each other. A solo model trained four times longer also stays at 0.0% (its best logged accuracy at any step averages 1.8%). **H1 is not supported in the pre-registered regime.** Interaction did change the models' internal similarity even without improving accuracy: linear CKA between the two models on test inputs rose from 0.33 (solo) to 0.68 (mutual), and prediction disagreement fell from 99.8% to about 80%. The comparisons against the big pooled model are formally "real" (mutual is 93.2 points lower) but uninformative, since nothing was learned.

4.2 Near-Threshold Regime: a Large Effect (Post Hoc)

At 30% labeled per model (regime T, random split, 6 seeds), a single model starts to generalize (32.7%, range 23.7–42.6%). Table 3 and Figure 1 show the results. Mutual distillation reaches 99.7% (uniform), 99.9% (agreement-weighted) and 99.9% (challenge-weighted). The gain over solo is $+67.0$ points (95% CI $[+62.3, +71.5]$) and over the non-interacting ensemble $+56.5$ points ($[+48.5, +63.4]$). The self-teacher control reaches only 2.0%: distilling from a lagged copy of itself reinforces the model's own wrong predictions, so seeing the inputs in U is not enough by itself. Pooled labels reach 100% for both model sizes; the gaps between the mutual variants and the big pooled model are under 0.4 points (below the 3-point threshold), so interaction *ties* pooled labels here and does not beat them. Figure 2 shows the trajectories: the mutual variants rise well before solo does, and solo only climbs late and is unstable at the end of training.

Table 3: Regime T (30% labeled per model), random split, 6 seeds. Test accuracy at the final step, mean with per-seed range.

Condition	Test accuracy	Range
solo (mean of A, B)	32.7%	23.7–42.6%
ensemble of solo A + B	43.1%	29.7–59.7%
self-teacher (control)	2.0%	0.9–2.6%
mutual, uniform	99.7%	99.0–100.0%
mutual, agreement-weighted	99.9%	99.9–100.0%
mutual, challenge-weighted	99.9%	99.3–100.0%
pooled labels, small	100.0%	100.0–100.0%
pooled labels, big	100.0%	100.0–100.0%

4.3 Controls

A different data history is required. When both models receive the same labeled set and run the identical mutual procedure, accuracy is 21.3%, no better than the 27.5% they reach without interacting. With different labeled sets it is 99.7%; the difference is $+78.3$ points ($[+75.0, +81.4]$). Having two models, or seeing U, is not what matters; what matters is that the two models hold different information to exchange.

It is not extra training. Training a solo model four times longer (24000 steps) ends at 22.1% (range 16.0–27.5%); its best accuracy at any logged step averages 33.1%. Mutual distillation reaches 99.7% in 6000 steps, a gap of $+77.6$ points ($[+74.5, +80.7]$) over the longer solo run.

Weighting. Challenge-weighted and agreement-weighted learning are indistinguishable from uniform and from each other in regime T (gaps of $+0.2$ points, $[-0.3, +0.6]$, challenge vs. uniform, and -0.1, $[-0.3, +0.1]$, challenge vs. agreement), largely because performance is at the ceiling. **H2 is not supported.**

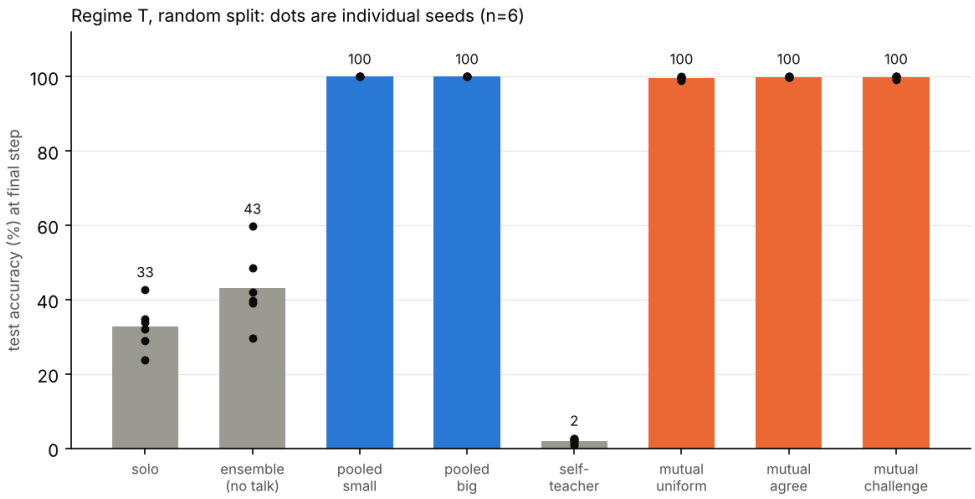


Figure 1: Regime T, random split: test accuracy at the final step. Bars are means; dots are individual seeds (6 seeds).

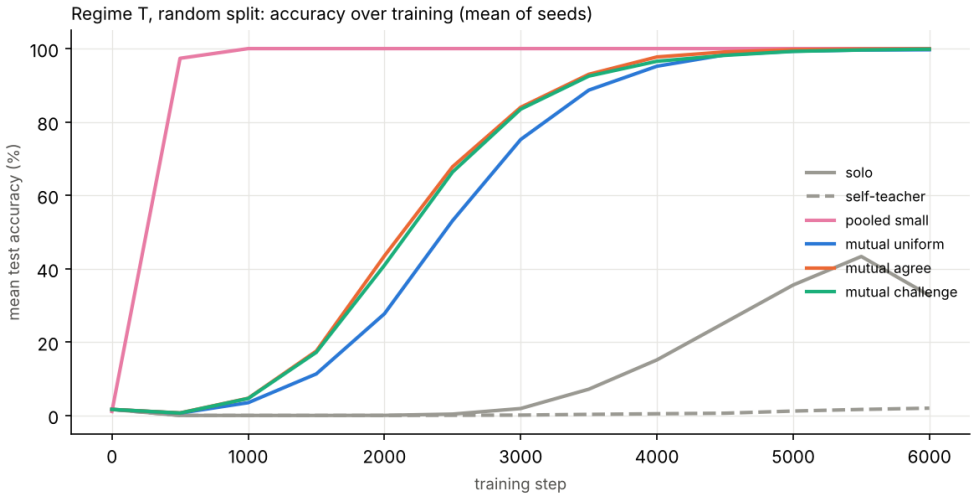


Figure 2: Regime T, random split: mean test accuracy over training.

4.4 Region Split: Retention and Transfer

In the region split (A sees labels only for $a < 27$, B only for $a \geq 27$) a solo model is 95.2% accurate in its own region and 0.0% in the other. After mutual distillation both models reach about 100% in both regions (other-region accuracy 99.6% uniform, 100.0% agreement-weighted, 99.9% challenge-weighted). Own-region accuracy rose by $+4.8$ points ($[+2.3, +8.1]$) rather than falling, so **H5 holds**: interaction did not damage each model's own domain. The non-interacting ensemble is already strong in this setting (92.0%, range 75.6–99.1%), so interaction adds less here, but mutual distillation still beats it by $+7.9$ points ($[+2.8, +14.7]$). The self-teacher control lowers own-region accuracy to 86.7% (-8.5 points, $[-12.6, -4.4]$).

4.5 Labeled-Fraction Sweep: the Effect Is a Band

To see where the effect lives, we swept the labeled fraction per model from 20% to 34% (post hoc, new seeds 10–12, 3 seeds per point, regime-T interaction settings without re-tuning; Table 4,

Figure 3). Below about 22% nothing is exchanged. Between 24% and 28% the mutual variants jump to 68–98% while solo is still at 1–7%: at 26% labeled, uniform mutual is at 67.6% and the weighted variants at 79.1% and 79.5%, against 1.2% for solo. From 30% up, pooled labels already suffice and interaction ties them. Pooled labels match or beat interaction at every point. Where the variants are not at the ceiling (26%), both weighted versions sit about 11–12 points above uniform, with overlapping per-seed ranges and 3 seeds; agreement-weighting is as good as challenge-weighting. This is suggestive that *weighting* may help, not that *challenge* specifically does. Note also that the uniform variant was tuned with a different λ than the weighted ones.

Table 4: Labeled-fraction sweep (post hoc; 3 seeds per row, seeds 10–12; regime-T settings). Mean test accuracy at the final step.

Labeled / model	solo	mutual, uniform	mutual, agreement	mutual, challenge	pooled (small)
20%	0.0%	0.0%	0.1%	0.1%	95.0%
22%	0.0%	0.4%	0.8%	0.6%	98.0%
24%	0.1%	8.2%	6.8%	8.3%	100.0%
26%	1.2%	67.6%	79.1%	79.5%	100.0%
28%	6.8%	95.2%	98.1%	97.4%	100.0%
30%	26.0%	100.0%	99.9%	99.9%	100.0%
32%	47.0%	100.0%	99.9%	99.8%	100.0%
34%	60.8%	100.0%	100.0%	100.0%	100.0%

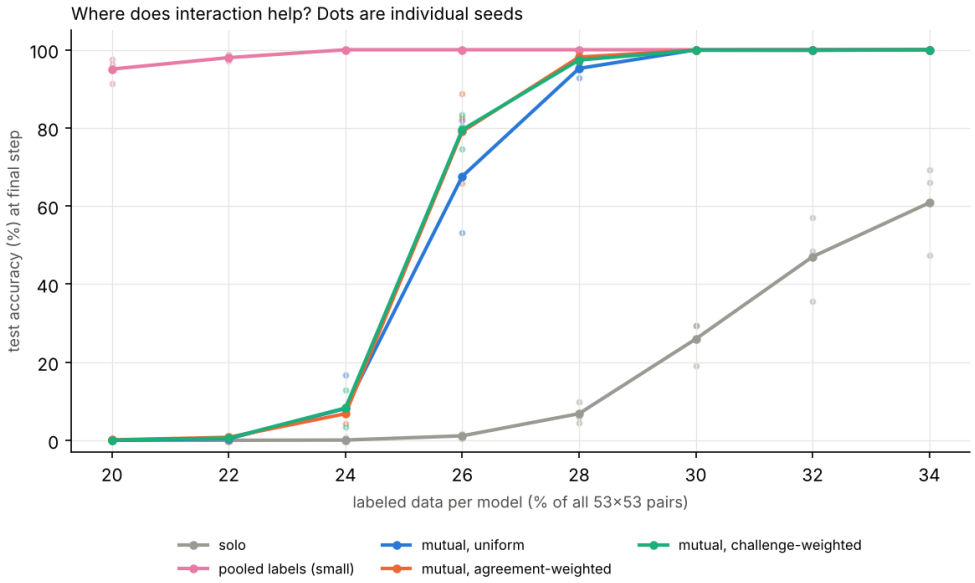


Figure 3: Test accuracy as a function of the labeled fraction per model (post hoc; 3 seeds per point). Lines are means; faint dots are individual seeds.

4.6 Diversity

Interaction reduces the difference between the two models (H4). In the random split of regime T, prediction disagreement on test inputs falls from 91.0% (solo) to 0.4% (uniform mutual). This is largely because both models are now nearly always right, so it is not by itself evidence of lost representational diversity. Linear CKA between the two models' hidden representations [Kornblith et al., 2019] moves in opposite directions in the two split modes: it rises from 0.39 to 0.57 in the random split and falls from 0.76 to 0.44 in the region split, so H4 is mixed on the representational measure. The two models end up making nearly identical predictions, so what interaction produces here is transfer of knowledge, not lasting diversity.

5 Discussion

In this toy task, two partial learners with different labeled data can push each other across a generalization threshold that neither crosses alone, using only each other's predictions on unlabeled inputs. Three observations support reading this as genuine exchange. The gain disappears when both models have the same data, it is far larger than what longer training or an ensemble of non-interacting models gives, and distilling from a lagged copy of oneself does not help. A plausible account is that each model's predictions on U act as pseudo-labels that carry information about the *other* model's labeled slice; we did not isolate the mechanism further, so this is a hypothesis.

The result should be read with its structure in mind. The pre-registered prediction failed: in regime P there is nothing to exchange. The positive result is in a regime chosen after that failure, with interaction settings tuned there. The sweep shows that this is not a robust general effect but a band of data amounts just below the point where a single model begins to generalize, and the sharp threshold of modular addition probably flatters it. Pooled labels were never beaten (H3 ties at best), so the finding concerns moving knowledge between models through unlabeled inputs, not producing more than the data contain. The mechanism is the known one of mutual learning, and we claim no novelty for it.

6 Limitations

One toy task with a sharp threshold. Modular addition has a grokking-style transition that produces large effects over a narrow range of data. Results may not transfer to tasks without such a threshold.

Few seeds. Regime P has 4 seeds, regime T 6, and the sweep 3 per point. The bootstrap intervals are crude, and the sweep edges are approximate.

Post-hoc positive result. The regime in which interaction helps was chosen after the pre-registered regime failed, and the interaction hyperparameters were tuned there. A confirmatory test would fix the regime and settings in advance and use new seeds.

Per-condition tuning. Distillation settings were tuned separately for each condition, so differences between mutual variants are confounded with tuning, and the weighting question is unresolved.

Narrow channel. Interaction is logit-level exchange on shared unlabeled inputs. Latent-space interaction, weight updates during interaction, and different architectures were not tested.

Same task for both models. Both models solve the same task, so this tests different data histories, not different skills. Compositional settings, where one model learns one skill and the other a different one, were not tested.

Ceiling effects. Many conditions reach 100%, which hides differences between variants.

Code availability. All code, raw results and analysis scripts are at `https://github.com/Malhar379/mutual-revision-partial-learners`.

7 Future Work

The most useful next steps are, in order: more seeds in the 24–28% band, with interaction settings retuned per point, to pin down the band's edges and test whether weighting really beats uniform; a task without a ceiling or a sharp threshold, so that weighting variants can be separated; a compositional-skills task in which one model learns one skill and the other a different one and the test needs both; comparison against named co-training and deep-mutual-learning baselines; and only then small language models and channels richer than output probabilities.

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